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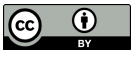
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Controlled Excitation for Identification of Nonlinear Systems with Saturated Outputs *

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Abstract

We consider nonlinear systems whose measured output is the product of an unknown matrix and a known saturation function of the unmeasured state. Generating excitation that allows the matrix to be recovered from the output is challenging when the dynamics are largely unknown. We exploit saturation to construct this excitation. Under a geometric accessibility condition, we derive explicit open-loop controls that drive the state into regions where the saturation takes prescribed values. For componentwise saturations, this condition holds generically with as few as two independent control directions. The controls require no state measurement and only partial knowledge of the dynamics, with no information on the structure of its nonlinear part. The resulting regressor is known in advance, and the unknown matrix is recovered exactly by a closed-form regression. Under stronger structural assumptions and with sufficiently many independent outputs, an output feedback tracks prescribed reference trajectories within the unsaturated region to arbitrary accuracy and permits exact state recovery. Suitably chosen references then also enable exact identification of linearly parameterized (nonlinear) dynamics. We analyze robustness to measurement noise and illustrate the method numerically in the presence of both measurement and dynamical noise.

1 Introduction

Saturations and other sigmoidal activation functions occur in many controlled systems. In neuroscience they are part of the standard models of population activity, such as the Wilson–Cowan equations [46, 45, 23], Amari neural fields [1] and neural mass models [30, 22]. In these models the saturation converts the mean membrane potential of a population into a firing rate [19, 11]. Although typical saturations are smooth, piecewise linear saturations, reaching their bounds at finite levels, are often used in practice since they make the models more tractable [20]. We refer to [10] for a review of such models at the scale of the whole brain.

Control theory has been applied to these models for brain monitoring [41], brain–machine interfaces [42] and closed-loop deep brain stimulation in Parkinson's disease [16, 24]. All of these applications need a model whose parameters are known. Fitting the parameters offline runs into degeneracy, since distinct parameter sets can reproduce the same data equally well

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[8], and it cannot follow parameters that drift over time [42]. In such models, moreover, the recorded signals are often weighted sums of the activity of several populations, which depend on the recording device and on the anatomy: this paper is about the identification of such weights.

More precisely, in this work we consider input–output systems of the form

$$\dot{x} = f(x, u, t), \qquad y = WS(x), \tag{1}$$

where the dynamics of the state $x \in \mathbb{R}^n$ is governed by the control $u \in \mathbb{R}^m$ via the map f, which is largely unknown; the output $y \in \mathbb{R}^q$ is measured (but not the full state x); S is a known map $\mathbb{R}^n \to \mathbb{R}^n$ assumed constant on some regions (we call such a map a saturation function, see Definition 3); and W is an unknown constant $q \times n$ matrix, whose determination from the measurement of y on a finite time interval is one of the main goals of the paper. We stress that the control u is left free, and a particularity of our construction is to design a control to ease identification.

This problem is an example of parameter identification [33, 39]. For controlled systems, it can usually be solved by a combination of two ingredients: a persistent excitation condition [7], and an observer or a contraction property of the dynamics [25, 9] in order to fit the input-output mapping to the observed data. See, e.g., [13] for a refined example of such principles in the context of dynamics with saturation. These tools, however, do not apply to (1).

In this formulation, the unknown matrix W appears multiplied by the time-varying signal $S(x(\cdot))$, that cannot in general play the role of a regressor[1]; it is indeed not measured and cannot be computed, since neither x nor f is fully known, and thus no excitation condition on it can be checked a priori, or even formulated in terms of the data. Observers do not help, as the contraction, passivity or observability properties they need are not available for f in general. Relaxations of persistent excitation, such as excitation over a finite interval [32], initial excitation [37], concurrent learning [18] and dynamic regressor extension and mixing [5], are not available. They weaken the requirement on the regressor, but still need it to be known or computable along the trajectory. The same holds for input design methods for nonlinear identification [44, 35], which assume that the regressor, or at least its structure, is available.

This is the typical situation of the high-uncertainty models of neuroscience, in which only the structure of the equations is known. State and parameter estimation for neural models has been addressed with Kalman filtering [38, 27], with observers for neural mass models [17, 36, 34], and with adaptive observers for networks and neural fields [12, 14, 15, 13, 3, 6]. These methods require either a detailed model of the dynamics or contraction and persistent excitation assumptions. Such assumptions are hard to check, and they are only weakly satisfied when saturation decouples some of the variables [2]. Observe that this question also arises for artificial neural networks, whose weight matrices one may wish to reconstruct from input–output data [4].

We propose an approach that exploits the specific features of saturation nonlinearities. Indeed, a saturation is constant on large regions of the state space, so the value of the regressor $S(x(\cdot))$ is known whenever the state is known to lie in one of these regions. In particular, if the control can steer in open loop the state into these regions (say, where $S(x) = v_i$ for known vectors v_i) and keep it there for some time, the output is $y = Wv_i$ on known intervals. Then, W follows from a linear regression whose regressor is known before the experiment is

[1] By regressor we intend, following [33, Sec. 7.3], the quantity φ appearing in an equality of the type $y(t) = \varphi(t)\theta$, on which many common identification methods are based.

run. In this sense the excitation is *designed* rather than assumed, and the dynamics enters only through the requirement that the saturation regions are reachable. The paper turns this idea into three results.

A general identification principle An excitation design is a finite list of saturation values $v_1, \dots, v_N$ spanning $\mathbb{R}^n$, together with disjoint time intervals $J_1, \dots, J_N$. A trajectory realizes the design if $S(x(t)) = v_i$ on each J_i (Definition 4). Proposition 5 shows that along any such trajectory an explicit closed-form estimator returns W exactly. This principles naturally doesn't depend on the dynamics f; neither does it require an observer. It is also general in terms of rank of W.

Excitation by open-loop controls We then give conditions under which a design can be realized. Namely, we consider dynamics of the form

$$\dot{x} = Ax + \phi(x, u, t) + Bu,$$

where A is a perturbation of a Hurwitz matrix A_0 and ϕ an unknown bounded perturbation. This class contains the Wilson–Cowan networks mentioned above. The saturation regions are assumed to contain coradiant sets (i.e. stable by dilation of factor larger than 1), as they do for componentwise saturations (see Definition 8). We assume that the image of $A_0^{-1}B$ intersects each saturation region, which holds for generic A_0 for componentwise saturations as soon as rank $B \geq 2$ (see Proposition 11). Under this condition, Theorems 9 and 12 provide explicit controls, computed from the known data alone, such that every solution starting in a prescribed ball realizes an excitation design. The controls are uniform with respect to the initial condition and to the unknown perturbation, within known bounds. The first theorem only needs the linear part up to a small perturbation, and the second one gives the cases in which the design can be realized in arbitrarily short time. Based on the previous identification principle, these theorems construct an open-loop experiment that identifies the matrix W, under mild structural assumptions on the dynamics and the saturations. This is in a sense the main result of the paper. We then pursue with reconstruction of the state and additional parameters, under significantly stronger assumptions.

Output feedback tracking Once W is known, the problem can be reversed. Instead of driving the state into the saturation regions, the control may be used to keep it away from them, in the region where the output is an invertible function of the state, if the saturations are of a more special —but usual— kind, and if the matrix B is invertible (see Section 2.3). Under it, Theorem 14 constructs an output feedback, with a sufficiently large gain, that brings every trajectory into the unsaturated region in prescribed time and tracks a reference curve there, so that the state is recovered by componentwise inversion of the saturation.

Together, the three results give a three-stage procedure, stated in Section 2.4: An open-loop experiment identifies W, an output feedback gives access to the state, and, for Wilson–Cowan networks, a reference chosen to excite the dynamics yields the coupling matrix by a second closed-form regression (Theorem 16).

Remark 1. This paper extends [2], where we considered a layered network of Wilson–Cowan type in which a single population is fully measured, and the interconnection weights were identified layer by layer by an online recursive procedure. Here, on the other hand, the dynamics is unknown apart from its linear part, the considered saturations are more general

than the classical componentwise one, and identification relies on a single open-loop experiment and a closed-form estimator.

Structure of the paper Section 2 presents the main results. Section 3 contains the geometric lemmas on coradiant sets, the construction of the controls and the proofs of the identification theorems. Section 4 is devoted to the tracking by output feedback, which yields the state, including the effect of measurement noise. Section 5 illustrates the identification procedure on a two-dimensional network with noise.

Notation

We let $[\![a, b]\!]$ be the set of integers between a and b, $\mathring{K}$ be the topological interior of a set K, and $\mathcal{B}(v, \rho)$ be the closed ball of radius ρ centered at $v \in \mathbb{R}^n$. We denote by $\|\cdot\|$ both the Euclidean norm on $\mathbb{R}^n$ and the associated operator norm (spectral norm) for matrices. We also denote the characteristic function of a set K by $\mathbf{1}_K$, and the Moore–Penrose pseudoinverse of a matrix M by M^+.

2 Main results

Our results, as stated in this section, get progressively stronger or more precise under progressively more restrictive assumptions on the dynamics (f) and the saturation function S in system (1). As already mentioned, our main case of interest is that of Wilson–Cowan networks, which involve a more particular form of f (see Section 2.4), and functions S that are componentwise saturation in the following sense.

Definition 2. A continuous function $S : \mathbb{R}^n \to \mathbb{R}^n$ is a *componentwise saturation* if $S(x) = (S_0(x_1), \dots, S_0(x_n))^\top$ with $S_0 : \mathbb{R} \to \mathbb{R}$ nondecreasing and such that, for some $\underline{R} < \bar{R}$, S_0 is strictly increasing on $[\underline{R}, \bar{R}]$ and

$$S_0(\xi) = 0 \qquad \text{if } \xi \le \underline{R}, \qquad \text{and} \qquad S_0(\xi) = 1 \qquad \text{if } \xi \ge \bar{R}.$$

Although our assumptions always include Wilson–Cowan networks, we will, consider more general saturation types than above (see Definitions 3 and 8) highlighting the structural features underlying our results.

2.1 A general identification result

We first state a preliminary identification result that holds for system (1) in a fairly general setting. Concerning the vector field f, we only need it to be (for instance) continuous w.r.t. (x, u) and measurable w.r.t. t, so that solutions on time interval $[0, T]$, to be taken in the sense of Carathéodory [26], make sense in Definition 4 and Proposition 5.

For the immediate discussion, we only require the following broad definition of saturation.

Definition 3. A continuous function $S : \mathbb{R}^n \to \mathbb{R}^n$ is a *saturation function* if there exist a family of closed sets with nonempty interior $K_1, \dots, K_N \subset \mathbb{R}^n$ and vectors $v_1, \dots, v_N \in \mathbb{R}^n$, such that

$$S(x) = v_i, \qquad \forall x \in K_i,\ \forall i \in [\![1, N]\!]. \tag{2}$$

Each set K_i is a *saturation region* of S and v_i is its *saturation value*.

A componentwise saturation in the sense of Definition 2 is naturally a saturation function: just take $I = \{0,1\}^n$, $v_i = i$, $i \in I$, and, with $R = \max\{|\underline{R}|, |\bar{R}|\}$,

$$K_v := \{x \in \mathcal{O}_v \mid |x_j| \geq R, j \in [\![1,n]\!]\},$$
$$\text{with } \mathcal{O}_v := \{x \in \mathbb{R}^n \mid x_j \geq 0 \text{ if } v_j = 1,\ x_j \leq 0 \text{ if } v_j = 0\} \quad (3)$$

(i.e. $\mathcal{O}_v$ is the closed orthant carrying the pattern v).

When the structure of the saturation is sufficiently rich, i.e., when the saturation values span $\mathbb{R}^n$, the unknown matrix W can be recovered exactly from the output of a trajectory that visits the saturation regions in a prescribed order and for prescribed durations. This is codified in the following definition, which is the main assumption of the identification result.

Definition 4. An *excitation design* for the saturation function S is a collection $\mathcal{E} = \{(v_i, J_i) \mid i \in I\}$, where $I \subset [\![1,N]\!]$ is a finite index set which up to relabeling we take as $I = [\![1,M]\!]$, the v_i are saturation values of S, and the $J_i \subset \mathbb{R}$ are pairwise disjoint closed intervals of positive length. Finally, we require that

$$\operatorname{rank}[v_1, \ldots, v_M] = n. \quad (4)$$

An absolutely continuous trajectory $x : [0,T] \to \mathbb{R}^n$ is a *realization of $\mathcal{E}$ in time $T > 0$* if $J_i \subset [0,T]$ for every i and

$$S(x(t)) = v_i \qquad \text{for all } t \in J_i \text{ and all } i \in [\![1,M]\!]. \quad (5)$$

We say that x is *saturation-exciting* if it realizes an excitation design.

Finally, the excitation design $\mathcal{E}$ is said to be *uniformly realizable in time $T > 0$ and radius $\rho > 0$* if there exists a control $u : [0,T] \to \mathbb{R}^m$ such that any solution of (1) under this control u with initial condition $\|x(0)\| \leq \rho$ is defined on $[0,T]$ and is a realization of $\mathcal{E}$ in time T.

Section 2.2 provides sufficient conditions for the existence of a uniformly realizable excitation design.

Given an excitation design $\mathcal{E}$, write $\tau_i := |J_i| > 0$ for the length of the i-th interval and define the excitation Gramian and the estimator

$$\Gamma(\mathcal{E}) := \sum_{i=1}^{M} \tau_i\, v_i v_i^\top \in \mathbb{R}^{n\times n}, \quad E(y;\mathcal{E}) := \left(\sum_{i=1}^{M} \int_{J_i} y(s)\, v_i^\top\, ds\right) \Gamma(\mathcal{E})^{-1} \in \mathbb{R}^{q\times n}. \quad (6)$$

The following observation is at the core of our identification procedure.

Proposition 5. *Let $\mathcal{E}$ be an excitation design and let (x,u,y) be a solution of (1) on $[0,T]$ whose state realizes $\mathcal{E}$. Then $\Gamma(\mathcal{E})$ is invertible and*

$$E(y;\mathcal{E}) = W. \quad (7)$$

Proof. The matrix $\Gamma(\mathcal{E})$ is symmetric and positive semidefinite. Indeed, for any $z \in \mathbb{R}^n$, we have $z^\top \Gamma(\mathcal{E}) z = \sum_i \tau_i (v_i^\top z)^2 \geq 0$. Moreover, $z^\top \Gamma(\mathcal{E}) z = 0$ forces $v_i^\top z = 0$ for every i. By (4), this implies that $z = 0$. To conclude, observe that by (5) the output verifies $y(t) = WS(x(t)) = Wv_i$ for every $t \in J_i$. Therefore

$$\sum_{i=1}^{N} \int_{J_i} y(s) v_i^\top\, ds = \sum_{i=1}^{N} \tau_i\, W v_i v_i^\top = W\, \Gamma(\mathcal{E}). \quad (8)$$

Multiplying on the right by $\Gamma(\mathcal{E})^{-1}$ yields the result. □

A recursive least-squares implementation of this regression is possible, see [28] or [33, Sec. 7.3].

Remark 6. The hypotheses of Proposition 5 can be relaxed in two ways. First, suppose the values of the design span only a proper subspace $V := \operatorname{span}\{v_1, \dots, v_M\}$ of $\mathbb{R}^n$, so that (4) fails. The Gramian $\Gamma(\mathcal{E})$ is then singular with range V, but the data still determine part of W. Indeed, replacing $\Gamma(\mathcal{E})^{-1}$ with the Moore–Penrose pseudoinverse in the definition of $E(y;\mathcal{E})$, the above argument yields

$$E(y;\mathcal{E}) = WP_V, \tag{9}$$

where P_V is the orthogonal projector onto V. Second, the equality of the source and target dimensions of S is not used in the proof. If $S : \mathbb{R}^n \to \mathbb{R}^d$ and $W \in \mathbb{R}^{q\times d}$, it suffices that the saturation values of the visited regions span $\mathbb{R}^d$.

Remark 7 (Measurement noise). Let the output be measured as $\tilde{y} = y + \nu$ along a trajectory realizing $\mathcal{E}$. For open-loop controls (such as those of Section 2.2), a bounded noise ν does not affect the realization, and the linearity of (6) then yields

$$\|E(\tilde{y};\mathcal{E}) - W\| \le \|\nu\|_\infty \Big(\sum_{i=1}^M \tau_i \|v_i\|\Big) \|\Gamma(\mathcal{E})^{-1}\| \le \|\nu\|_\infty \Big(\sum_{i=1}^M \tau_i \|v_i\|\Big) \frac{\|(\mathcal{V}\mathcal{V}^\top)^{-1}\|}{\min_i \tau_i},$$

where we let $\mathcal{V} := [v_1, \dots, v_M]$.

2.2 Realizing excitation designs

Proposition 5 is of use only if an excitation design can be produced. Here, we consider the following specialization of (1),

$$\begin{cases} \dot{x} = Ax + \phi(x,u,t) + Bu, \\ y = WS(x). \end{cases} \tag{10}$$

We assume that

- A is a (small) perturbation of a known Hurwitz matrix A_0, i.e.,

  $$A = A_0 + A_1 \tag{11}$$

  with A_1 unknown and satisfying $\|A_1\| \le \bar{a}$ for some known $\bar{a} \ge 0$,
- ϕ is an unknown continuous and bounded function, with $\|\phi\|_\infty \le \bar{\phi}$ for some known $\bar{\phi} > 0$, and
- $B \in \mathbb{R}^{n\times m}$ is known.

This setting includes Wilson–Cowan networks with an unknown coupling matrix [46, 20], that we discuss in Section 2.4. As for the function S, we now on require it to be a coradiant saturation in the following sense.

Definition 8. The function S is a *coradiant saturation* if there exist n closed coradiant sets[2] $K_1, \dots, K_n \subset \mathbb{R}^n$ with non-empty interior, and linearly independent vectors $v_1, \dots, v_n \in \mathbb{R}^n$ such that

$$S(x) = v_i, \quad \forall x \in K_i, \quad i \in [\![1, n]\!]. \tag{12}$$

[2]Recall that a set $K \subset \mathbb{R}^n$ is coradiant if $\lambda x \in K$ for all $x \in K$ and $\lambda \ge 1$.

Coradiant saturations are clearly saturations in the sense of Definition 3. Componentwise saturations (Definition 3) are coradiant saturations because the sets K_v in (3) are coradiant, and $\{0,1\}^n$ contains the canonical basis of $\mathbb{R}^n$.

The following result is proved in Section 3.3, based on the beginning of Section 3.

Theorem 9. *Let S be a coradiant saturation, and assume that*

$$\mathrm{Im}(A_0^{-1}B) \cap \mathring{K}_i \neq \varnothing \qquad \forall i \in [\![1,n]\!]. \tag{13}$$

Then, there exists $\varepsilon_0 > 0$ with the following property: for any $\bar{a} < \varepsilon_0$, there exists $T^* \geq 0$ such that for every $\rho > 0$ and every $T > T^*$ there exists an excitation design $\mathcal{E}$ which is uniformly realizable in time T and radius ρ for all A_1 with $\|A_1\| \leq \bar{a}$ and all ϕ with $\|\phi\|_\infty \leq \bar{\phi}$, via a control that does not depend on them. Moreover, there is $\bar{\rho} \geq \rho$, independent of A_1 and ϕ, such that these solutions satisfy $\|x(t)\| \leq \bar{\rho}$ for every $t \in [0,T]$.

Remark 10. In the above result, ε_0 depends only on A_0, B and $K_1, \ldots, K_n$, while T^* depends also on $\bar{a}$ and is given by (55). The control and the radius $\bar{\rho}$ are explicit. The control is piecewise constant, and both depend on the known data alone, namely A_0, B, $K_1, \ldots, K_n$, $\bar{a}$, $\bar{\phi}$, ρ and T.

Condition (13) is always satisfied if $\mathrm{rank}\, B \geq n$ and never if $\mathrm{rank}\, B = 1$ and $n \geq 3$. Depending on the saturation, this condition can be weakened: for componentwise saturations we have the following proposition, proved at the end of Section 3.3.

Proposition 11. *Let S be a componentwise saturation. Then, for any B with* $\mathrm{rank}\, B \geq 2$, *the set of Hurwitz matrices A_0 such that condition* (13) *holds for suitable sets $K_{\mathbf{s}}$ of* (3) *is open and dense in the set of Hurwitz matrices.*

The idea underlying the construction in Theorem 9 is the fact that, for a linear system $\dot{x} = A_0 x + Bu$ with A_0 Hurwitz, for any $b \in \mathrm{Im}(B)$ one can steer the system towards $-\mu A_0^{-1} b$ using the constant controls $Bu \equiv \mu b$. In particular, under (13), taking μ sufficiently large allows to sequentially drive the state to the desired regions $\mathring{K}_1, \ldots, \mathring{K}_n$. Making this procedure insensitive to the perturbation A_1 requires then that it be sufficiently small, while the bounded term ϕ is dominated by taking μ large.

Under additional assumptions on the unperturbed part A_0, the design can be realized in arbitrarily short time.

Theorem 12. *Under the assumptions of Theorem 9 hold. assume that*

i. either $\bar{a} = 0$, that is, $A = A_0$ is known, and the pair (A_0, B) is controllable,

ii. or $A_0 = -\lambda\,\mathrm{Id}$ for some $\lambda > 0$.

Then, the conclusion of Theorem 9 holds with $T^* = 0$, that is, for every $T > 0$.

Remark 13. Also in this case the control and the radius $\bar{\rho}$ are explicit and depend on the known data alone, namely A_0 (or λ), B, $K_1, \ldots, K_n$, $\bar{a}$, $\bar{\phi}$, ρ and T. The control is piecewise constant in case *ii.*, while in case *i.* each step uses control (39).

2.3 Tracking by output feedback and state recovery

We now assume that the matrix W is known, e.g. by implementing Proposition 5 and Theorem 9, and proceed with state reconstruction, under three more assumptions. First, we suppose that B in (10) is the identity matrix[3], and consider the following further specialization of (10):

$$\begin{cases} \dot{x} = Ax + \phi(x,u,t) + u, \\ y = WS(x). \end{cases} \tag{14}$$

Second, we assume that S *is a componentwise saturation*[4] rather than merely a coradiant saturation; this adds invertibilty properties outside saturation regions: S is injective on $\Omega := [\underline{R}, \bar{R}]^n$. Third, we assume that the *known matrix* W *is left invertible* (i.e., $\operatorname{rank} W = n$; this requires $q \geq n$); then, fixing a left inverse H of W, the second line in (14) reads

$$Hy = S(x). \tag{15}$$

The following result, proved in Section 4, allows tracking of any reference γ staying in $\mathring{\Omega}$, via an explicit output feedback law; since S is invertible on $\mathring{\Omega}$, the state can then be read off the output from (15).

Theorem 14. *Let the previous assumptions hold. Let $\rho > 0$ and $\varepsilon > 0$, let $0 < T_1 < T_2$, and let γ be a C^1 curve with bounded derivative taking its values in a compact subset $\mathcal{K}$ of $\mathring{\Omega}$. Then there exist $k^\star > 0$ and continuous maps $\eta_k : \mathbb{R}^q \times \mathbb{R}^n \to \mathbb{R}^m$, $k > 0$, such that, for every $k > k^\star$, every solution of (14), with $\|x(0)\| \leq \rho$, under the output feedback $u(t) = \eta_k(y(t), \gamma(t))$ is defined for all positive times and satisfies*

$$x(t) \in \mathring{\Omega} \quad \forall t \geq T_1 \qquad \text{and} \qquad \|x(t) - \gamma(t)\| \leq \varepsilon \quad \forall t \geq T_2, \tag{16}$$

and $x_j(t) = S_0^{-1}\big((Hy(t))_j\big)$ for $t \geq T_1$ and $j \in [\![1,n]\!]$, where $S_0^{-1} : [0,1] \to [\underline{R}, \bar{R}]$ is the inverse of $S_0|_{[\underline{R},\bar{R}]}$.

Moreover, if S_0^{-1} is locally Lipschitz on $(0,1)$, there exist $\delta_0, c_\nu > 0$ such that for every measurable $\nu : [0,+\infty) \to \mathbb{R}^q$ with $\|\nu(t)\|_\infty \leq \delta \leq \delta_0$, and the output feedback $u(t) = \eta_k(y(t) + \nu(t), \gamma(t))$, the same conclusions as above hold with $\varepsilon + c_\nu\delta$ in place of ε in (16). Also, the recovered state $\hat{x}_j(t) := S_0^{-1}\big((H(y(t)+\nu(t)))_j\big)$ is well defined for $t \geq T_1$ and satisfies $\|\hat{x}(t) - x(t)\|_\infty \leq c_\nu\delta$.

Remark 15. In the above result, the maps η_k and the constants $k^\star$, δ_0 and c_ν are explicit and given in Section 4. In particular, the maps η_k depend only on W, S, S_0 and $\mathcal{K}$, while $k^\star$, which is defined in (67), does not depend on W. None depends on ν, and c_ν does not depend on k, so the floor $c_\nu\delta$ is not lowered by a larger gain.

2.4 Identification of the full system

We conclude this section by showing how to combine the identification of W and the output feedback of Theorem 14 to recover the state, and then the dynamics, from the output. We

[3] If B is invertible, we can always reduce to this case by a change of variables $u \mapsto Bu$.

[4] If, instead, $S = PS_1$ with P known and invertible and S_1 a componentwise saturation, then the second line in (14) reads $y = (WP)S_1(x)$, and everything below applies with WP in place of W.

showcase this procedure in the case of Wilson–Cowan networks [46, 20], where the nonlinear part of the dynamics (14) is given by

$$\phi(x,u,t) = C\sigma(x), \tag{17}$$

with σ a known continuous bounded function (for instance, a saturation, unrelated to S) and C an unknown constant matrix.

Theorem 16. *Consider* (14) *with S a componentwise saturation and W injective, and assume that $\bar{a} < \varepsilon_0$, where ε_0 and T^* are given by Theorem 9. Let $\rho > 0$, $T > T^*$, $\varepsilon > 0$ and $0 < T_1 < T_2$, and let γ and $\mathcal{K}$ be as in Theorem 14. Then there exist an excitation design $\mathcal{E}$, a constant $k^\star > 0$ and, for every $k > k^\star$, a control u, open loop on $[0,T)$ and an output feedback on $[T,+\infty)$, both independent of A_1, ϕ, and without prior knowledge of W, such that every solution with $\|x(0)\| \le \rho$ satisfies the following.*

i. $\hat{W} := E(y|_{[0,T]}; \mathcal{E}) = W$.

ii. $x(t) \in \mathring{\Omega}$ *and* $x_j(t) = S_0^{-1}\big((\hat{W}^+ y(t))_j\big)$ *for* $t \ge T + T_1$ *and* $j \in [\![1,n]\!]$, *and* $\|x(t) - \gamma(t - T)\| \le \varepsilon$ *for* $t \ge T + T_2$.

iii. Assume moreover $A = A_0$ is known and $\phi(x,u,t) = C\sigma(x)$, with $C \in \mathbb{R}^{n\times n}$ unknown and $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ known, continuous and bounded.

Assume there exist $p^1, \dots, p^n \in \mathcal{K}$ such that $P := [\sigma(p^1), \dots, \sigma(p^n)]$ is invertible. Pick γ such that $\gamma(t) = p^\ell$ on disjoint intervals $t \in I_\ell = [a_\ell, b_\ell] \subset [T_2, \infty)$, $a_\ell < b_\ell$, for all $\ell \in [\![1,n]\!]$, and let

$$d_\ell := x(T + b_\ell) - x(T + a_\ell) - \int_{T+I_\ell} (Ax + u)\, dt, \quad \zeta_\ell := \int_{T+I_\ell} \sigma(x)\, dt. \tag{18}$$

Then there exists $\varepsilon_1 > 0$, depending only on σ and $p^1, \dots, p^n$, such that if $\varepsilon < \varepsilon_1$, the matrix $Z := [\zeta_1, \dots, \zeta_n]$ is invertible and $C = [d_1, \dots, d_n]\, Z^{-1}$.

Proof. The proof is divided into three parts, corresponding to the three items of the statement.

Step 1: Identification of W. As observed in Section 2.2, S is a coradiant saturation with the sets K_{e_j} of (3) and $v_j = e_j$. Since $B = \mathrm{Id}$, condition (13) holds. Hence Theorem 9 gives a control u_{ol} on $[0,T]$, a design $\mathcal{E}$ and a radius $\bar{\rho}$, independent of A_1 and ϕ, such that every solution with $\|x(0)\| \le \rho$ realizes $\mathcal{E}$ and satisfies $\|x(T)\| \le \bar{\rho}$. Proposition 5 then gives item *i*.

Step 2: Tracking. Let $\hat{\eta}_k$ be the feedback of Theorem 14 built with $\hat{W}$ and the left inverse $H = \hat{W}^+$, and consider

$$u(t) = \begin{cases} u_{\mathrm{ol}}(t), & t \in [0,T), \\ \hat{\eta}_k\big(y(t), \gamma(t-T)\big), & t \ge T. \end{cases} \tag{19}$$

Since $x(T + \cdot)$ solves (14) with ϕ shifted in time, which obeys the same bound $\bar{\phi}$, and $\|A\| \le \|A_0\| + \bar{a}$ is known, the conclusions of Theorem 14, applied with $\bar{\rho}$ in place of ρ, hold for $k > k^\star$ with t replaced by $t - T$. This is item *ii*.

Step 3: Identification of C. By item *ii*., x is known on $T + I_\ell \subset [T + T_1, +\infty)$, so d_ℓ and ζ_ℓ are computed from y and u. Integrating the dynamics $\dot{x} = Ax + C\sigma(x) + u$ over $T + I_\ell$ gives $d_\ell = C\zeta_\ell$, so it suffices to show that Z is invertible. Let ω_σ be a modulus of continuity of σ on Ω. Since $\|x(T+t) - p^\ell\| \le \varepsilon$ for $t \in I_\ell$, we have $\|\zeta_\ell - |I_\ell|\, \sigma(p^\ell)\| \le |I_\ell|\, \omega_\sigma(\varepsilon)$. Thus, $Z = (P + R)\,\mathrm{diag}(|I_1|, \dots, |I_n|)$ with $\|R\| \le \sqrt{n}\, \omega_\sigma(\varepsilon)$. In particular, Z is invertible as soon as $\sqrt{n}\, \omega_\sigma(\varepsilon) < \|P^{-1}\|^{-1}$, which defines ε_1. □

Other identification procedures, such as those based on persistent excitation, can also be used at this stage. Accordingly, this procedure can be extended to more general ϕ, as soon as the corresponding dynamics are identifiable by prescribing the state and knowledge of the input.

With the noisy output of Theorem 14, the estimator of item *iii.* is evaluated on the recovered state $\hat{x}$, which is within $c_\nu\delta$ of x, while u is known exactly, since it is the applied control. Hence, if σ is Lipschitz and $\varepsilon + c_\nu\delta < \varepsilon_1$, the error on C is of order δ. More windows can also be combined by least squares, as in Section 5.

Remark 17. If A is also unknown, the same argument applies to $[A\ C]$ with the regressor $(x, \sigma(x)) \in \mathbb{R}^{2n}$ and $2n$ points p^ℓ. Such points exist if and only if the components of x and $\sigma(x)$ are linearly independent functions on $\mathring{\Omega}$, which holds for a componentwise σ if and only if its profile σ_0 is not affine on $[\underline{R}, \bar{R}]$.

3 Realization of the excitation designs

In this section we prove Theorems 9 and 12, which assume the saturation to be coradiant, as per Definition 8. Hence, the natural choice for the vectors of the design $\mathcal{E}$ is $v_1, \dots, v_n$. Realizing such a design consists in bringing the state into K_1 and holding it there during the interval J_1, then into K_2 during J_2, and so, on up to K_n. Thus, we need to do two things: *(i)* construct a control bringing the state into one prescribed region and keeping it there, and *(ii)* chain n such controls.

Section 3.1 addresses the first ingredient. It measures how much room a coradiant region leaves around a given direction, and then solves the one-region problem. Section 3.2 chains the steps, and Section 3.3 proves the theorems.

3.1 Steering into one region

We introduce the following notion.

Definition 18. Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior and let $T > 0$. We say that (10) *can be steered into* K *in time* T if for every $\rho > 0$ and every initial time $t_0 \geq 0$ there exist a bounded measurable control $u : [t_0, +\infty) \to \mathbb{R}^m$ and a radius $\rho' \geq \rho$ such that every solution of (10) with $\|x(t_0)\| \leq \rho$ satisfies

$$x(t) \in K \quad \forall t \geq t_0 + T, \qquad \|x(t)\| \leq \rho' \quad \forall t \geq t_0. \tag{20}$$

Observe that the same T must work for every initial ball, but the control and the radius ρ' may depend on ρ. For coradiant saturations and for $K = K_i$, (20) implies $S(x(t)) = v_i$ for every $t \geq t_0 + T$.

In this section we present three lemmas, each of which establishes that (10) can be steered into a coradiant region in time T, strictly positive for the first one and arbitrarily small for the second and third. To state them, we need to introduce a measure of how much room a coradiant region leaves around a given direction.

3.1.1 Margin of a coradiant region

Definition 19. Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, and let $v \in \mathring{K}$, $v \neq 0$. The *margin* of v in K is

$$\delta_K(v) := \min\left\{1,\ \frac{\operatorname{dist}(v, \mathbb{R}^n \setminus K)}{\|v\|}\right\} \in (0,1]. \tag{21}$$

The margin measures the distance from v to the boundary of K, relative to $\|v\|$. On a convex cone it is the sine of the angle from v to the nearest face. On an orthant it is $\min_j |v_j|/\|v\|$.

The steering lemmas of Section 3.1 are stated in terms of the following quantity, which depends only on the data of the problem, and not on a particular direction.

Definition 20. Let A' be Hurwitz and let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior. The *best margin* of K with respect to A' is

$$\delta_K^*(A') := \sup\big\{\delta_K(w) :\ w \in \operatorname{Im}(A'^{-1}B) \cap \mathring{K}\big\} \ \in\ [0,1], \tag{22}$$

with the convention $\sup \varnothing = 0$.

Observe that $\delta_K^*(A') > 0$ is equivalent to $\operatorname{Im}(A'^{-1}B) \cap \mathring{K} \neq \varnothing$, and that every margin below $\delta_K^*(A')$ is exceeded by the margin of some $w \in \operatorname{Im}(A'^{-1}B) \cap \mathring{K}$.

Remark 21. The supremum in (22) is in general not a maximum, and closedness of K does not help. For the componentwise saturations the margin increases strictly along each ray, toward the limit $\min_j |w_j|/\|w\|$, which is therefore never attained.

The following comparison is the core of the three steering lemmas below. Figure 1 shows why the margin must exceed the residual.

Lemma 22. *Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, let $w \in \mathring{K}$ with $w \neq 0$, and let $\delta \in (0, \delta_K(w)]$. Let $s_0 \geq 1$, $c_1 \geq 0$ and $c_2 \geq 0$ satisfy*

$$s_0\,(\delta - c_1)\,\|w\| \ \geq\ c_2. \tag{23}$$

Then, for every $s \geq s_0$ and every $r \in \mathbb{R}^n$ such that

$$\|r\| \ \leq\ s\,c_1\|w\| + c_2, \tag{24}$$

we have $sw + r \in K$.

Proof. Since $c_2 \geq 0$ and $s_0\|w\| > 0$, condition (23) forces $c_1 \leq \delta$. Using $s \geq s_0$, (24) and then (23),

$$s\delta\|w\| - \|r\| \ \geq\ s(\delta - c_1)\|w\| - c_2 \ \geq\ s_0(\delta - c_1)\|w\| - c_2 \ \geq\ 0. \tag{25}$$

In particular $\|r\| \leq s\delta\|w\|$. Denote by $\mathcal{B}(v,\rho)$ the closed ball of radius ρ centered at v. Since $\delta \leq \delta_K(w)$ and K is closed, the definition of the margin gives $\mathcal{B}(w, \delta\|w\|) \subset K$. Since $s \geq 1$ and K is coradiant, it follows that

$$\mathcal{B}(sw, s\delta\|w\|) \ =\ s\,\mathcal{B}(w, \delta\|w\|) \ \subset\ sK \ \subset\ K,$$

and therefore $sw + r \in K$. □

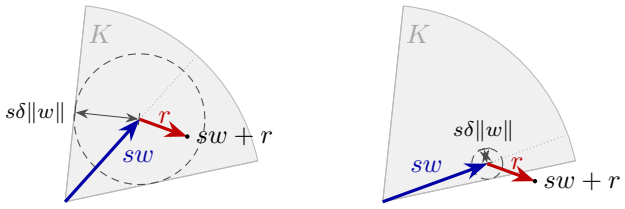


Figure 1: Lemma 22: K contains the ball of radius $s\delta\|w\|$ centered at sw (dashed). Left: the margin absorbs r. Right: with a small margin, the same r leaves K.

3.1.2 The steering lemmas

For simplicity we present all statements and proofs with $t_0 = 0$, the general case being obtained by a time translation. In the three lemmas below, the direction b and the threshold μ_0, hence the control, are computed from A_0, B, K, T, ρ, $\bar{\phi}$ and $\bar{a}$ only, and not from A_1 or ϕ.

Since A_0 is Hurwitz, there exist constants $M_0 \geq 1$ and $\alpha_0 > 0$ such that

$$\|e^{A_0 t}\| \leq M_0 e^{-\alpha_0 t}, \qquad t \geq 0. \tag{26}$$

Once a solution x and a control u are fixed, we abbreviate $\phi(s) := \phi(x(s), u(s), s)$. In particular, the solution x is given by the variation of constants formula

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-s)}\big[\phi(s) + Bu(s)\big]ds, \qquad t \geq 0. \tag{27}$$

Our first steering lemma uses a constant control, which forces a lower bound on the time needed to steer the system into the coradiant region. The control is built from A_0, while the equilibrium it reaches, $-A^{-1}b$, involves the unknown A_1. Recall that a bound $\bar{a} \geq \|A_1\|$ is known, and write

$$p := \frac{M_0}{\alpha_0}\,\bar{a}, \qquad \vartheta := \frac{p}{1-p} \quad (p < 1), \qquad \alpha_1 := \alpha_0 - M_0\bar{a}. \tag{28}$$

Since $\|A_0^{-1}\| \leq M_0/\alpha_0$ by (26), we have $\|A_0^{-1}A_1\| \leq p$, and ϑ bounds how far A_1 can move that equilibrium relative to $\|A_0^{-1}b\|$.

Lemma 23 (Steering by a constant control)**.** *Let A_0 be Hurwitz with constants M_0, α_0 as in (26), and let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior. Set $\delta^* := \delta_K^*(A_0)$ and assume that*

$$\bar{a} \;<\; \frac{\alpha_0}{M_0} \cdot \frac{\delta^*}{1+\delta^*}. \tag{29}$$

With ϑ and α_1 as in (28), set

$$T_{\min} := \frac{1}{\alpha_1} \ln \frac{M_0}{\hat{\delta}}, \quad \text{where} \quad \hat{\delta} := \frac{\delta^* - \vartheta}{1+\vartheta}. \tag{30}$$

Then, (10) can be steered into K in every time $T > T_{\min}$. More precisely, given $T > T_{\min}$, there exists $b \in \mathrm{Im}(B)$ such that, for every $\rho > 0$, there exists $\mu_0 \geq 1$ such that, for every $\mu \geq \mu_0$, the constant control $Bu \equiv \mu b$ drives every solution of (10) with $\|x(0)\| \leq \rho$ so that (20) holds with $\rho' = R_\mu^0$ given by

$$R_\mu^0 := \mu(1+M_0)(1+\vartheta)\|A_0^{-1}b\| + M_0\rho + \frac{M_0\bar{\phi}}{\alpha_1}. \tag{31}$$

Proof. We start by estimating the decay of e^{At}. Recall that

$$e^{At} = e^{A_0 t} + \int_0^t e^{A_0(t-s)} A_1 e^{As} ds.$$

Hence, the bound (26) and $\|A_1\| \le \bar{a}$ show that $\psi(t) \le M_0 + M_0 \bar{a} \int_0^t \psi(s) ds$, for $\psi(t) := e^{\alpha_0 t}\|e^{At}\|$. Grönwall's lemma gives

$$\|e^{At}\| \;\le\; M_0 e^{-\alpha_1 t}, \qquad t \ge 0. \tag{32}$$

By (29), $M_0\bar{a} < \alpha_0$, so $\alpha_1 > 0$, and in particular A is Hurwitz.

Let now $b \in \mathrm{Im}(B)$ be such that $w_0 := -A_0^{-1} b \in \mathring{K}$ and

$$M_0 e^{-\alpha_1 T} \;<\; \hat{\delta}_0 := \frac{\delta_0 - \vartheta}{1+\vartheta}, \qquad \text{where } \delta_0 := \delta_K(w_0). \tag{33}$$

Such a b exists by definition of the best margin, since $\hat{\delta}_0 \to \hat{\delta}$ as $\delta_0 \to \delta^*$. In the following, we first determine the location of the equilibrium $w := -A^{-1}b$ actually reached by the control with respect to the boundary of K. Then, we consider the trajectory of (10) under the constant control $Bu \equiv \mu b = -\mu A w$, and conclude by applying Lemma 22 with μ sufficiently large.

Let us show that $w \in \mathring{K}$ and estimate its margin. Consider $P := A_0^{-1} A_1$, so that $A = A_0(\mathrm{Id} + P)$ and $\|P\| \le p$. By (28) and (29), $p < \delta^*/(1+\delta^*) \le 1/2$. In particular, $\mathrm{Id} + P$ is invertible by the Neumann series [29], and $w = (\mathrm{Id}+P)^{-1} w_0$. Moreover

$$\|w - w_0\| = \big\|(\mathrm{Id}+P)^{-1} P w_0\big\| \;\le\; \frac{\|P\|}{1-\|P\|}\|w_0\| \;\le\; \vartheta \|w_0\|, \tag{34}$$

so that $(1-\vartheta)\|w_0\| \le \|w\| \le (1+\vartheta)\|w_0\|$. Since the left-hand side of (33) is positive, we have $\hat{\delta}_0 > 0$, that is, $\vartheta < \delta_0$. It follows that $w \in \mathcal{B}(w_0, \delta_0 \|w_0\|)$, which is contained in K by definition of δ_0. Hence,

$$\mathrm{dist}\big(w, \mathbb{R}^n \setminus K\big) \;\ge\; (\delta_0 - \vartheta)\|w_0\| \;>\; 0.$$

In particular, $w \in \mathring{K}$. Dividing by $\|w\| \le (1+\vartheta)\|w_0\|$ and using $\delta_0 \le 1$ gives

$$\delta_K(w) \;\ge\; \hat{\delta}_0 \;\in\; (0, 1]. \tag{35}$$

Consider now the trajectory x of (10) under the constant control $Bu \equiv \mu b = -\mu A w$. The variation of constants formula (27) gives

$$x(t) = \mu w + \varrho(t), \qquad \varrho(t) := e^{At}\big(x(0) - \mu w\big) + \int_0^t e^{A(t-s)} \phi(s)\, ds, \qquad t \ge 0. \tag{36}$$

By (32), $\|x(0)\| \le \rho$ and $\|\phi\| \le \bar{\phi}$, for every $t \ge T$,

$$\|\varrho(t)\| \;\le\; \mu M_0 e^{-\alpha_1 T}\|w\| + M_0 \rho + \frac{M_0 \bar{\phi}}{\alpha_1}. \tag{37}$$

Let us now apply Lemma 22 at each $t \ge T$, with $s = s_0 = \mu$, $r = \varrho(t)$, $\delta = \hat{\delta}_0$, $c_1 = M_0 e^{-\alpha_1 T}$ and $c_2 = M_0\rho + M_0\bar{\phi}/\alpha_1$. The choices for δ and r are admissible thanks to (35) and (37).

Condition (33) is exactly $c_1 < \hat{\delta}_0$. In order to apply Lemma 22, it remains to check (23), which is implied by $\mu \geq \mu_0$, where

$$\mu_0 := \max\left\{1,\ \frac{M_0\rho + M_0\bar{\phi}/\alpha_1}{\big(\hat{\delta}_0 - M_0 e^{-\alpha_1 T}\big)(1-\vartheta)\|w_0\|}\right\}. \tag{38}$$

Indeed, since $\mu \geq \mu_0 \geq 1$ and $\|w\| \geq (1-\vartheta)\|w_0\|$, we have

$$\mu\big(\hat{\delta}_0 - c_1\big)\|w\| \ \geq\ \mu\big(\hat{\delta}_0 - c_1\big)(1-\vartheta)\|w_0\| \ \geq\ c_2.$$

Lemma 22 therefore gives the first part of (20). Finally, (36) and (32) give

$$\|x(t)\| \ \leq\ \mu\|w\| + M_0\big(\rho + \mu\|w\|\big) + \frac{M_0\bar{\phi}}{\alpha_1} \ \leq\ R^0_\mu, \qquad \forall t \geq 0,$$

using $\|w\| \leq (1+\vartheta)\|w_0\|$ once more. Since $R^0_\mu \geq M_0\rho \geq \rho$, the system can be steered into K in time T in the sense of Definition 18, with $\rho' = R^0_\mu$. □

We now show that adding some assumptions on the linear part allows one to steer in arbitrary time. The first case we consider is when A is known and the pair (A, B) is controllable. Here, given an amplitude $\mu > 0$ and a direction b with equilibrium $w = -A^{-1}b$, we steer to μw at time T and hold the state there, using

$$u(t) = \begin{cases} \mu\, B^\top e^{A^\top(T-t)}\mathcal{G}(T)^{-1}w, & t \in [0,T), \\ u_b, & t \geq T, \end{cases} \tag{39}$$

where $u_b \in \mathbb{R}^m$ is any vector with $Bu_b = \mu b$, and $\mathcal{G}(T)$ is the controllability Gramian,

$$\mathcal{G}(T) := \int_0^T e^{As} B B^\top e^{A^\top s}\, ds. \tag{40}$$

Since (A, B) is controllable, $\mathcal{G}(T)$ is invertible and (39) steers the system to μw at time T [40, 43, 21]. Associated with the Gramian is the constant

$$c_B(T) := \sup_{0 \leq t \leq T} \big\| \mathcal{G}(t)\, e^{A^\top(T-t)}\mathcal{G}(T)^{-1} \big\|. \tag{41}$$

That is, $c_B(T)$ is the maximal gain along the minimum-energy trajectory steering the origin to a unit vector in time T. It is finite by continuity of $t \mapsto \mathcal{G}(t)e^{A^\top(T-t)}$.

Our result is then the following.

Lemma 24 (Steering in arbitrary time, controllable case)**.** *Let $A_1 = 0$, so that $A = A_0$ is Hurwitz, assume that (A, B) is controllable, let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, and assume $\delta^*_K(A) > 0$. Then* (10) *can be steered into K in* every *time $T > 0$. More precisely, there exists $b \in \mathrm{Im}(B)$ such that, for every $T > 0$ and every $\rho > 0$, there exists $\mu_0 \geq 1$ such that, for every $\mu \geq \mu_0$, the control* (39) *drives every solution of* (10) *with $\|x(0)\| \leq \rho$ so that* (20) *holds with*

$$\rho' := \mu\big(1 + c_B(T)\big)\|A^{-1}b\| + (1 + M_0)\left(M_0\rho + \frac{M_0\bar{\phi}}{\alpha_0}\right). \tag{42}$$

Proof. Let $b \in \mathrm{Im}(B)$ be such that $w := -A^{-1}b \in \mathring{K}$, which exists since $\delta_K^*(A) > 0$, and set $\delta := \delta_K(w)$. The open-loop part of (39) places the state at μw at time T, up to the contributions of the initial condition and of ϕ. More precisely, at $t = T$ the variation of constants formula (27) yields $\|x(T) - \mu w\| \le M_0\rho + M_0\bar{\phi}/\alpha_0$. For $t \ge T$ the control is the constant $Bu \equiv \mu b$, so, similarly to (36), we have

$$x(t) = \mu w + \varrho(t), \qquad \varrho(t) = e^{A(t-T)}(x(T) - \mu w) + \int_T^t e^{A(t-s)}\phi(s)\,ds, \qquad t \ge T, \tag{43}$$

hence

$$\|\varrho(t)\| \;\le\; (1 + M_0)\left(M_0\rho + \frac{M_0\bar{\phi}}{\alpha_0}\right), \qquad t \ge T. \tag{44}$$

This is the form (24) required by Lemma 22, with $s = s_0 = \mu$, $r = \varrho(t)$, $\delta = \delta_K(w)$, $c_1 = 0$ and c_2 the right-hand side of (44). Condition (23), together with $s_0 \ge 1$, then holds for every $\mu \ge \mu_0$, where

$$\mu_0 := \max\left\{1,\; \frac{(1 + M_0)\big(M_0\rho + M_0\bar{\phi}/\alpha_0\big)}{\delta\,\|w\|}\right\}. \tag{45}$$

Lemma 22, applied at each $t \ge T$, therefore gives $x(t) \in K$ for all $t \ge T$. As for the state bound, on $[0, T]$ Duhamel's formula and (39) give $\|x(t)\| \le M_0\rho + M_0\bar{\phi}/\alpha_0 + \mu c_B(T)\|w\|$, where $c_B(T)$ is defined in (41), while on $[T, +\infty)$ we have $\|x(t)\| \le \mu\|w\| + \|\varrho(t)\|$. Both are at most ρ'. □

When the linear part is not fully known, the minimal time can be removed if the known part is $A_0 = -\lambda\,\mathrm{Id}$ for some $\lambda > 0$, although the conditions on the unknown part are stronger than in Lemma 23. We observe that in this case, $M_0 = 1$ and $\alpha_0 = \lambda$, so that $p = \bar{a}/\lambda$ and $\alpha_1 = \lambda - \bar{a}$ in (28).

Lemma 25 (Steering in arbitrary time, scalar known part)**.** *Let $A_0 = -\lambda\,\mathrm{Id}$ with $\lambda > 0$. Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior. Set $\delta^* := \delta_K^*(A_0)$ and assume that*

$$\bar{a} \;<\; \lambda \cdot \frac{\delta^*}{2 + \delta^*}. \tag{46}$$

Then (10) *can be steered into K in* every *time $T > 0$. More precisely, there exists $b \in \mathrm{Im}(B)$ such that, for every $T > 0$ and every $\rho > 0$, there exists $\mu_0 \ge 1$ such that, for every $\mu \ge \mu_0$, the constant control $Bu \equiv \mu b$ drives every solution of* (10) *with $\|x(0)\| \le \rho$ so that* (20) *holds with $\rho' = R^0_\mu$ defined in* (31)*.*

Proof. Let $b \in \mathrm{Im}(B)$ be such that $w_0 := b/\lambda \in \mathring{K}$ and $\delta_0 := \delta_K(w_0) > 2\vartheta$. Such a b exists by definition of the best margin, since (46) is equivalent to $2\vartheta < \delta^*$. By (46) we also have $\bar{a} < \lambda\delta^*/(1 + \delta^*)$, which is the hypothesis (29) of Lemma 23. Since moreover $\vartheta < \delta_0/2 < \delta_0$, the same argument as in the proof of the latter places $w := -A^{-1}b$ in $\mathring{K}$, with

$$\delta_K(w) \;\ge\; \hat{\delta}_0 = \frac{\delta_0 - \vartheta}{1 + \vartheta}, \qquad \|w\| \;\ge\; (1 - \vartheta)\frac{\|b\|}{\lambda}. \tag{47}$$

Moreover $\vartheta < 1/2$ gives $p < 1/3$, hence $\alpha_1 = \lambda(1 - p) > 2\lambda/3$.

Since $A_0 = -\lambda \operatorname{Id}$ commutes with A_1, we have $e^{At} = e^{-\lambda t} e^{A_1 t}$, so that $\|e^{At}\| \le e^{-(\lambda - \|A_1\|)t} \le e^{-\alpha_1 t}$. Under the constant control $Bu \equiv \mu b$, the variation of constants formula (27) and $\int_0^t e^{A(t-s)} ds = A^{-1}(e^{At} - \operatorname{Id})$ give, for any $t \ge 0$,

$$x(t) = \mu\big(\operatorname{Id} - e^{At}\big)w + \varrho(t), \quad \varrho(t) := e^{At}x(0) + \int_0^t e^{A(t-s)}\phi(s)\, ds. \tag{48}$$

In particular, the residual ϱ is bounded uniformly in t and in μ by $\|\varrho(t)\| \le \rho + \bar{\phi}/\alpha_1$.

It therefore suffices to keep the curve $t \mapsto (\operatorname{Id} - e^{At})w$ far enough inside K for every $t \ge T$. To this end, we split the curve into a part along the ray $\mathbb{R}_+ w$ and a remainder $z(t)$, as follows:

$$\big(\operatorname{Id} - e^{At}\big)w = \big(1 - e^{-\lambda t}\big)w + z(t), \qquad z(t) := e^{-\lambda t}\big(\operatorname{Id} - e^{A_1 t}\big)w. \tag{49}$$

From $\|\operatorname{Id} - e^{A_1 t}\| \le e^{\|A_1\| t} - 1 \le e^{\bar{a}t} - 1$ and $\bar{a} < \lambda$, we can bound the remainder for $t \ge 0$ as

$$\begin{aligned} \|z(t)\| \;&\le\; e^{-\lambda t}\big(e^{\bar{a}t} - 1\big)\|w\| \;=\; \bar{a}\|w\| \int_0^t e^{\bar{a}\tau - \lambda t}\, d\tau \\ &\le\; \bar{a}\|w\| \int_0^t e^{-\lambda(t-\tau)}\, d\tau \;=\; \big(1 - e^{-\lambda t}\big)\frac{\bar{a}}{\lambda}\,\|w\|. \end{aligned} \tag{50}$$

Both terms of (49) carry the same factor $1 - e^{-\lambda t}$, so their comparison is independent of t and of μ, and no minimal time is needed.

In order to conclude the proof, we now apply Lemma 22 to the trajectory (48). Let us check that the hypotheses are satisfied. Substituting (49) into (48) gives $x(t) = \mu\big(1 - e^{-\lambda t}\big)w + \big(\mu z(t) + \varrho(t)\big)$, which by (50) has the form (24) required by Lemma 22, with $s = \mu(1 - e^{-\lambda t}) \ge s_0 := \mu(1 - e^{-\lambda T})$, $r = \mu z(t) + \varrho(t)$, $c_1 = \bar{a}/\lambda = \vartheta/(1+\vartheta)$ and $c_2 = \rho + \bar{\phi}/\alpha_1 < \rho + 3\bar{\phi}/(2\lambda)$ for $t \ge T$, since $t \mapsto 1 - e^{-\lambda t}$ is increasing. Moreover, by (47) we can take $\delta = \hat{\delta}_0$. Recalling that $2\vartheta < \delta_0$, we have

$$\delta - c_1 \;\ge\; \frac{\delta_0 - \vartheta}{1+\vartheta} - \frac{\vartheta}{1+\vartheta} \;=\; \frac{\delta_0 - 2\vartheta}{1+\vartheta} \;>\; 0. \tag{51}$$

With $\|w\| \ge (1-\vartheta)\|b\|/\lambda$, the condition (23) is implied by

$$\mu\big(1 - e^{-\lambda T}\big)\frac{(\delta_0 - 2\vartheta)(1-\vartheta)}{1+\vartheta} \cdot \frac{\|b\|}{\lambda} \;\ge\; \rho + \frac{3\bar{\phi}}{2\lambda},$$

and the requirement $s_0 \ge 1$ is $\mu(1 - e^{-\lambda T}) \ge 1$. Both hold for every $\mu \ge \mu_0$, where

$$\mu_0 := \frac{1}{1 - e^{-\lambda T}} \max\left\{1,\; \frac{\lambda(1+\vartheta)\big(\rho + 3\bar{\phi}/(2\lambda)\big)}{(1-\vartheta)(\delta_0 - 2\vartheta)\,\|b\|}\right\}. \tag{52}$$

Finally, Lemma 22 applied at each $t \ge T$ with $s = \mu\big(1 - e^{-\lambda t}\big)$ gives $x(t) \in K$ for all $t \ge T$, and $\|x(t)\| \le R_\mu^0$ for every $t \ge 0$ follows from $\|e^{At}\| \le 1$ and $M_0 = 1$. □

3.2 Concatenating the steps

In this section we show that the previous results can be concatenated to sequentially steer the system into a sequence of coradiant sets.

Lemma 26. *Let S be a coradiant saturation. For each $i \in [\![1,n]\!]$, let (10) be steerable into K_i in some time $T_i > 0$ in the sense of Definition 18. Then for every $\rho > 0$, and $T, \tau > 0$ such that $T \geq n\tau + \sum_{i=1}^n T_i$, there exist intervals J_i with $|J_i| = \tau$ such that $\mathcal{E} = \{(v_i, J_i)\}_{i\in[\![1,n]\!]}$ is an excitation design uniformly realizable in time T and radius ρ. Moreover, there is $\bar\rho \geq \rho$ such that every solution of (10) with $\|x(0)\| \leq \rho$ under the control corresponding to $\mathcal{E}$ satisfies $\|x(t)\| \leq \bar\rho$ for every $t \in [0,T]$.*

Proof. Set $s_0 := 0$ and define the times

$$t_1 := T_1, \qquad t_i := s_{i-1} + T_i, \qquad s_i := t_i + \tau, \quad i \in [\![1,n]\!], \tag{53}$$

and $J_i := [t_i, s_i]$, so that $t_1 \geq T_1$, $t_i \geq s_{i-1} + T_i$ and $s_n = n\tau + \sum_{i=1}^n T_i \leq T$. These intervals are pairwise disjoint, and the v_i are linearly independent since S is a coradiant saturation, so that (4) holds and $\mathcal{E} := \{(v_i, J_i)\}_{i\in[\![1,n]\!]}$ is an excitation design. The construction of the control is by induction on i, the induction hypothesis being a bound ρ_i on the norm of the state at time s_{i-1}, with $\rho_1 := \rho$. Given ρ_i, Definition 18 applied with the initial time s_{i-1} furnishes for K_i and the radius ρ_i a control u_i and a radius ρ_i' such that any solution issued at time s_{i-1} from a state of norm at most ρ_i satisfies $x(t) \in K_i$ for every $t \geq s_{i-1} + T_i$, hence for every $t \in J_i$ by (53), and $\|x(t)\| \leq \rho_i'$ throughout. By (12), $S(x(t)) = v_i$ on J_i, which is the i-th requirement for the excitation design, see (5). Setting $\rho_{i+1} := \rho_i'$ closes the induction, and the concatenation $u := \sum_{i=1}^n u_i \mathbf{1}_{[s_{i-1}, s_i)}$, extended to $[0,T]$ by u_n on $[s_n, T]$, is the desired control. Finally, $\rho \leq \rho_1' \leq \cdots \leq \rho_n'$ by Definition 18, so $\bar\rho := \rho_n'$ bounds $\|x(t)\|$ on $[0,T]$. □

3.3 Proofs of the theorems

We are finally in a position to prove the main results of Section 2.2.

Proof of Theorem 9. For each i let $\delta_i := \delta^*_{K_i}(A_0) > 0$, and set

$$\varepsilon_0 := \frac{\alpha_0}{M_0} \min_{1\leq i\leq n} \frac{\delta_i}{1+\delta_i}. \tag{54}$$

Let $\bar a < \varepsilon_0$. Then $\bar a$ satisfies (29) with $\delta^* = \delta_i$ for every i. Lemma 23 therefore applies to each K_i, and makes (10) steerable into K_i in every time larger than

$$T^i_{\min} := \frac{1}{\alpha_1} \ln \frac{M_0}{\hat\delta_i}, \qquad \hat\delta_i := \frac{\delta_i - \vartheta}{1+\vartheta} \;>\; 0,$$

with ϑ and $\alpha_1 = \alpha_0 - M_0\bar a$ as in (28). Set

$$T^* := \sum_{i=1}^n T^i_{\min} = \sum_{i=1}^n \frac{1}{\alpha_0 - M_0\bar a} \ln \frac{M_0}{\hat\delta_i}. \tag{55}$$

Given $T > T^*$, we can choose times $T_i > T^i_{\min}$ with $\sum_i T_i < T$, and Lemma 26 with $\tau := (T - \sum_i T_i)/n > 0$ concludes. By Lemma 23, the directions, the times T_i and the amplitudes are all computed from A_0, B, the K_i, $\bar\phi$, ρ, T and $\bar a$. So are the intervals J_i and the radii ρ_i of the proof of Lemma 26, hence the design $\mathcal{E}$, the radius $\bar\rho$ and the control u, which is piecewise constant since each step uses a constant control. In particular, neither depends on A_1 or ϕ within the bounds $\|A_1\| \leq \bar a$ and $\|\phi\|_\infty \leq \bar\phi$. □

The time (55) increases with the bound $\bar{a}$ and blows up as $\bar{a} \to \varepsilon_0$. For $\bar{a} = 0$, it reduces to $\sum_{i=1}^{n} \alpha_0^{-1} \ln(M_0/\delta_i)$.

Proof of Theorem 12, case i. Here $\bar{a} = 0$, so $A_1 = 0$ and $A = A_0$ is known and Hurwitz and the hypothesis of Theorem 9 reads $\mathrm{Im}(A^{-1}B) \cap \mathring{K}_i \neq \varnothing$ for each i. Since (A, B) is controllable, Lemma 24 makes (10) steerable into K_i in *every* time $T_i > 0$. Given $T > 0$, taking the T_i small enough that $\sum_i T_i < T$ and applying Lemma 26 with $\tau := (T - \sum_i T_i)/n > 0$ concludes. Hence $T^* = 0$, and by Lemma 24 the control on the i-th step is computed from A_0, B, K_i, $\bar{\phi}$, ρ and T_i. □

Proof of Theorem 12, case ii. Here $A_0 = -\lambda \,\mathrm{Id}$ with $\lambda > 0$, so $M_0 = 1$, $\alpha_0 = \lambda$ and $A = -\lambda \,\mathrm{Id} + A_1$ has the structure required by Lemma 25. With $\delta_i := \delta^*_{K_i}(A_0) > 0$ as in the proof of Theorem 9, set

$$\varepsilon_0 := \lambda \min_{1 \leq i \leq n} \frac{\delta_i}{2 + \delta_i}, \tag{56}$$

which depends only on λ, B and the K_i. Since $\bar{a} < \varepsilon_0$, we have $\bar{a} < \lambda\delta_i/(2 + \delta_i)$ for every i, which is (46) for K_i. Lemma 25 then makes (10) steerable into K_i in every time $T_i > 0$, and one concludes as in case *i*. Hence $T^* = 0$, and by Lemma 25 the control on the i-th step is computed from λ, B, K_i, $\bar{\phi}$, ρ and $\bar{a}$. □

We conclude this section with the proof of Proposition 11.

Proof of Proposition 11. We first show openness. Let (13) hold for A_0 with sets $K_1, \ldots, K_n$, and pick $b_k \in \mathrm{Im}\, B$ with $A_0^{-1} b_k \in \mathring{K}_k$. Since $A_0' \mapsto A_0'^{-1} b_k$ is continuous near A_0 and $\mathring{K}_k$ is open, $A_0'^{-1} b_k \in \mathring{K}_k$ for every A_0' close enough to A_0, which can moreover be taken Hurwitz since being Hurwitz is an open condition.

We now show density, that is, that any Hurwitz A_0 has a Hurwitz A_0' arbitrarily close to it for which (13) holds. Let $b, b' \in \mathrm{Im}\, B$ be linearly independent, and set $Z_0 := A_0^{-1}[b\ b'] \in \mathbb{R}^{n \times 2}$, which has rank 2. For $Z \in \mathbb{R}^{n \times 2}$ with rows $\omega_1^\top, \ldots, \omega_n^\top$, let $p(Z) := \prod_{i<j} \det[\omega_i\ \omega_j]$. Then, p is a polynomial in the entries of Z which is non-zero. Indeed, $p(\bar{Z}) \neq 0$ for $\bar{Z}$ with rows $\bar{\omega}_i = (1, i)$. Since the zero set of a non-zero polynomial has empty interior, there exists Z arbitrarily close to Z_0 such that $p(Z) \neq 0$. In particular, the rows of Z are pairwise linearly independent, and thus $Z^\top Z$ is invertible. Fix any such Z and consider $A_0' := A_0 + A_0(Z_0 - Z)Z^\dagger$, with $Z^\dagger = (Z^\top Z)^{-1} Z^\top$. We have $A_0' Z = [b\ b']$ and $\|A_0' - A_0\| \leq \|A_0\|\, \|Z_0 - Z\|\, \|Z^\dagger\|$. As $Z^\dagger$ stays bounded near Z_0 and being Hurwitz is an open condition, it follows that A_0' is Hurwitz and arbitrarily close to A_0. In particular, A_0' is invertible and $\mathrm{Im}\, Z \subset \mathrm{Im}(A_0'^{-1} B)$.

Consider now the points Zy, whose coordinates are $x_i = \langle \omega_i, y \rangle$, for y on the unit circle of $\mathbb{R}^2$. Each x_i vanishes at exactly two antipodal points, and by the choice of Z these $2n$ points are distinct. They split the circle into $2n$ open arcs, on each of which no coordinate vanishes, so that Zy stays in the interior of a single orthant $\mathcal{O}_{\mathbf{s}}$. Crossing the zero of x_i changes the sign of x_i only, so that the patterns of two adjacent arcs differ by $\pm e_i$. Since every $i \in [\![1, n]\!]$ occurs, the span of these $2n$ patterns contains $e_1, \ldots, e_n$, and we can extract n linearly independent patterns $\mathbf{s}_1, \ldots, \mathbf{s}_n$. Finally, if y lies on an arc with pattern $\mathbf{s}_k$, then $\lambda Z y \in \mathring{K}_{\mathbf{s}_k}$ for λ large enough. Hence (13) holds for A_0' with $K_k := K_{\mathbf{s}_k}$. □

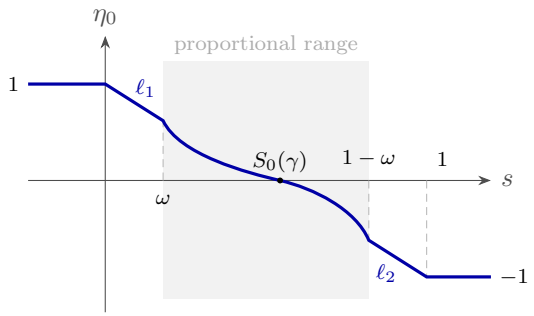


Figure 2: The scalar feedback (57) as a function of the measured coordinate s.

4 Output feedback tracking

In this section we prove the statements of Section 2.3, namely Theorem 14.

Observe that reading the output coordinate by coordinate tells whether each coordinate of the state is saturated. This suggests the following coordinate-wise strategy for the output feedback loop: In a saturated coordinate the law pushes the state back with a constant effort, while in an unsaturated one it switches to proportional tracking. The scalar feedback is obtained by scalar multiplication of the following function, where $\omega \in (0, 1/2)$ and $\gamma \in \mathbb{R}$:

$$\eta_0(s;\gamma,\omega) = \begin{cases} 1, & s \le 0, \\ \ell_1(s), & 0 < s < \omega, \\ -\big(S_0^{-1}(s) - \gamma\big), & \omega \le s \le 1-\omega, \\ \ell_2(s), & 1-\omega < s < 1, \\ -1, & s \ge 1, \end{cases} \tag{57}$$

where ℓ_1 and ℓ_2 are affine functions making η_0 continuous. Figure 2 shows the five branches. The actual output feedback control is then defined by a scalar gain $k > 0$ and a reference value $\gamma \in \mathbb{R}^n$ as

$$\eta_k(y,\gamma) := k\boldsymbol{\eta}_0\big(Hy;\gamma,\omega\big), \quad \boldsymbol{\eta}_0(s;\gamma,\omega) := \big(\eta_0(s_1;\gamma_1,\omega),\dots,\eta_0(s_n;\gamma_n,\omega)\big)^\top. \tag{58}$$

For $\omega \in (0, 1/2)$ write

$$\Omega_\omega := \big[S_0^{-1}(\omega),\, S_0^{-1}(1-\omega)\big]^n \subset \mathring{\Omega} \tag{59}$$

for the box obtained from Ω by staying at distance ω from the saturation levels, measured on the output side, and, for $L > 0$ and $X \subset \mathbb{R}^n$, let $C^{1,L}(\mathbb{R}, X)$ denote the set of C^1 curves with values in X whose derivative is bounded by L.

Lemma 27. *Let $g : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^n$ be continuous with*

$$\|g(x,u,t)\| \le a\|x\| + c_g, \qquad (x,u,t) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}, \tag{60}$$

for some known *constants $a, c_g > 0$, and consider*

$$\dot{x} = g(x,u,t) + u, \qquad y = WS(x), \tag{61}$$

with W known and injective, S a componentwise saturation and H as in (15). Let $\varepsilon, \rho, L > 0$, let $0 < T_1 < T_2$, let $\omega \in (0, 1/4)$ and let $\gamma \in C^{1,L}(\mathbb{R}, \Omega_{2\omega})$. Let $\nu : [0, +\infty) \to \mathbb{R}^q$ be measurable with $\|H\nu(t)\|_\infty \le \delta_H \le \omega/2$ for all t, where $\delta_H = 0$ unless S_0^{-1} is locally Lipschitz on $(0, 1)$. Then there exists $k^\star > 0$ such that for every gain $k > k^\star$ the solutions of (61) under the control $u(t) = \eta_k(y(t) + \nu(t), \gamma(t))$ of (58), with $\|x(0)\| \le \rho$, are global in time and satisfy

$$x(t) \in \Omega_\omega \quad \forall t \ge T_1, \qquad \|x(t) - \gamma(t)\| \le \varepsilon + \sqrt{n}\,\ell_\omega \delta_H \quad \forall t \ge T_2, \tag{62}$$

where ℓ_ω depends only on S and ω.

Proof. Continuity of S, g and η_0 and measurability of ν give local existence and absolute continuity of the solutions, and by (15) the closed loop reads

$$\dot{x}_j = g_j(x, u, t) + k\ \eta_0\big(S_0(x_j) + \tilde{\nu}_j(t); \gamma_j, \omega\big), \qquad j \in [\![1, n]\!], \quad \tilde{\nu} := H\nu. \tag{63}$$

Set

$$c_\omega := \min\Big\{1,\ S_0^{-1}(1 - \tfrac{3}{2}\omega) - S_0^{-1}(1 - 2\omega),\ S_0^{-1}(2\omega) - S_0^{-1}(\tfrac{3}{2}\omega)\Big\}\ >\ 0. \tag{64}$$

Observe that the positivity of c_ω follows from $\omega > 0$ and the strict monotonicity of S_0 on $[\underline{R}, \bar{R}]$, while $\omega < 1/4$ makes $\Omega_{2\omega}$ nonempty. For every t and $|r| \le \omega/2$, we claim that

$$\eta_0\big(S_0(\xi) + r; \gamma_j(t), \omega\big) \begin{cases} \le -c_\omega, & \xi \ge S_0^{-1}(1 - \omega), \\ \ge c_\omega, & \xi \le S_0^{-1}(\omega). \end{cases} \tag{65}$$

Indeed, recall that $\gamma_j(t) \in [S_0^{-1}(2\omega), S_0^{-1}(1 - 2\omega)]$, and let $\xi \ge S_0^{-1}(1 - \omega)$, so that $s := S_0(\xi) + r \ge 1 - \frac{3}{2}\omega > \omega$. If $s \le 1 - \omega$, then $\eta_0(s) = -(S_0^{-1}(s) - \gamma_j) \le -c_\omega$ by (64). Otherwise, η_0 is affine on $[1 - \omega, 1]$ and constant beyond, with $\eta_0(1 - \omega) \le -c_\omega$ and $\eta_0(1) = -1 \le -c_\omega$. The lower level is symmetric.

Consider now the following quantities

$$\Lambda := \max\big\{\rho, |\underline{R}|, |\bar{R}|\big\}, \qquad \kappa := a\sqrt{n}\,\Lambda + c_g + L, \tag{66}$$

$$k^\star := \max\left\{\frac{1}{c_\omega}\left(\frac{2\Lambda}{T_1} + a\sqrt{n}\,(\Lambda + 1) + c_g\right),\ \ \frac{2\sqrt{n}\,\kappa}{\varepsilon},\ \ \frac{1}{T_2 - T_1}\ln\frac{2\sqrt{n}\,(\bar{R} - \underline{R})}{\varepsilon}\right\}. \tag{67}$$

Throughout, $k > k^\star$, whose first term in particular gives

$$k\ >\ \frac{a\sqrt{n}\,(\Lambda + 1) + c_g}{c_\omega}. \tag{68}$$

We now deduce from (65) that $\{\|x\|_\infty \le \Lambda\}$ is positively invariant. Fix $c \in (0, 1]$, and let s_1 be the infimum of the times when $\|x(t)\|_\infty \ge \Lambda + c$. Assuming that s_1 is finite, there exists j such that $|x_j(s_1)| = \Lambda + c$, say $x_j(s_1) = \Lambda + c$, the other case being symmetric. Since $x_j(0) \le \rho \le \Lambda$, the set of times before s_1 with $x_j = \Lambda$ is nonempty and closed. Let s_0 be its maximum, so that $x_j > \Lambda$ on $(s_0, s_1]$. By (60) and the definition of Λ, for all $t \in [s_0, s_1]$,

$$|g_j(x(t), u(t), t)| \le a\sqrt{n}(\Lambda + c) + c_g \qquad \text{and} \qquad x_j(t) \ge \Lambda \ge \bar{R} \ge S_0^{-1}(1 - \omega). \tag{69}$$

Since $c \le 1$, using (63), (68), and (65) this gives $\dot{x}_j(t)\ \le\ a\sqrt{n}\,(\Lambda + c) + c_g - kc_\omega < 0$ almost everywhere on $[s_0, s_1]$, contradicting $x_j(s_1) > x_j(s_0)$. Hence $\|x(t)\|_\infty < \Lambda + c$ as long as the

solution is defined, so that the solution is bounded, hence global. Since $c \in (0,1]$ is arbitrary, we obtain $\|x(t)\|_\infty \le \Lambda$ for every $t \ge 0$, and therefore

$$\|g(x(t),u(t),t)\| \;\le\; a\sqrt{n}\,\Lambda + c_g, \qquad \forall t \ge 0. \tag{70}$$

We now show that the set Ω_ω is reached before T_1 and is invariant, which settles the first part of (62). By (65) and (70), whenever $x_j(t) \ge S_0^{-1}(1-\omega)$ one has $\dot{x}_j \le -(kc_\omega - a\sqrt{n}\Lambda - c_g) < 0$, and symmetrically $\dot{x}_j > 0$ whenever $x_j(t) \le S_0^{-1}(\omega)$. The interval $[S_0^{-1}(\omega), S_0^{-1}(1-\omega)]$ is therefore positively invariant for x_j, and since $|x_j(0)| \le \Lambda$ and $|S_0^{-1}(s)| \le \Lambda$ for $s \in [0,1]$, it is reached after a time at most $\frac{2\Lambda}{kc_\omega - a\sqrt{n}\,\Lambda - c_g}$, which tends to 0 as $k \to \infty$ and is at most T_1 by the first term of (67). Hence $x(t) \in \Omega_\omega$ for every $t \ge T_1$, the first assertion of (62).

Let now $t \ge T_1$. We have $S_0(x_j(t)) \in [\omega, 1-\omega]$, so $S_0^{-1}(S_0(x_j)) = x_j$ and (63) reads $\dot{x}_j = g_j - k(x_j - \gamma_j) + r_j$, with $r_j := k\big(\eta_0(S_0(x_j) + \tilde{\nu}_j) - \eta_0(S_0(x_j))\big)$. If S_0^{-1} is not locally Lipschitz on $(0,1)$, then $\delta_H = 0$, so that $r_j = 0$, and we set $\ell_\omega := 0$. Otherwise, let λ_ω be a Lipschitz constant of S_0^{-1} on $I_\omega := [\omega/2, 1-\omega/2]$. Then η_0 is λ_ω-Lipschitz on $[\omega, 1-\omega]$, and affine on $[\omega/2, \omega]$ and $[1-\omega, 1-\omega/2]$ with slope at most $(1 + \bar{R} - \underline{R})/\omega$ in absolute value, so that $|r_j| \le k\ell_\omega\delta_H$ with $\ell_\omega := \max\big\{\lambda_\omega,\ (1+\bar{R}-\underline{R})/\omega\big\}$. With $e_j := x_j - \gamma_j$ and $\|\dot{\gamma}\| \le L$, (70) and (66) give

$$D^+|e_j| \;\le\; \kappa + k\ell_\omega\delta_H - k|e_j| \qquad \text{almost everywhere on } [T_1, +\infty), \tag{71}$$

where D^+ is the upper right Dini derivative, $|e_j|$ being absolutely continuous but not differentiable at its zeros. Hence, by the comparison lemma [31] and $|e_j(T_1)| \le \bar{R} - \underline{R}$,

$$|e_j(t)| \;\le\; \big(\bar{R} - \underline{R}\big)e^{-k(t-T_1)} + \frac{\kappa}{k} + \ell_\omega\delta_H, \qquad t \ge T_1. \tag{72}$$

The constant κ does not depend on k, so the first two terms are at most $\varepsilon/(2\sqrt{n})$ for every $t \ge T_2$ as soon as

$$k \;>\; \max\left\{\frac{2\sqrt{n}\,\kappa}{\varepsilon},\ \frac{1}{T_2 - T_1}\ln\frac{2\sqrt{n}\,(\bar{R}-\underline{R})}{\varepsilon}\right\}, \tag{73}$$

whose right-hand side collects the two remaining terms of (67). It follows that $\|x(t) - \gamma(t)\| \le \sqrt{n}\max_j |e_j(t)| \le \varepsilon + \sqrt{n}\,\ell_\omega\delta_H$ for $t \ge T_2$, concluding the proof. □

We can now turn to the proof of Theorem 14.

Proof of Theorem 14. It suffices to prove the second part, since for $\nu = 0$ the argument below does not use the Lipschitz continuity of S_0^{-1} and gives the first part. Let $\mathcal{K} \subset \mathring{\Omega}$ be the compact set of the statement and let $L > 0$ bound $\|\dot{\gamma}\|$. Since $\mathcal{K}$ is compact and contained in the open box $\mathring{\Omega}$, there is $\zeta > 0$ with $\mathcal{K} \subset [\underline{R} + \zeta, \bar{R} - \zeta]^n$, and then, S_0^{-1} being continuous on $[0,1]$ with $S_0^{-1}(0) = \underline{R}$ and $S_0^{-1}(1) = \bar{R}$, there is some $\omega \in (0, 1/4)$ with $\mathcal{K} \subset \Omega_{2\omega}$, so that $\gamma \in C^{1,L}(\mathbb{R}, \Omega_{2\omega})$. The system (14) is of the form (61) with $g(x,u,t) = Ax + \phi(x,u,t)$, which is continuous and satisfies (60) with $a = \|A\| > 0$ and $c_g = \bar{\phi} > 0$, uniformly in u and t. We take for η_k the feedback (58) with this ω and with H as in (15). It depends only on k, W, S, S_0 and $\mathcal{K}$. Let $\|H\|_\infty$ be the operator norm of H for $\|\cdot\|_\infty$, and set $\delta_0 := \omega/(2\|H\|_\infty)$, so that $\|H\nu(t)\|_\infty \le \delta_H := \|H\|_\infty\delta \le \omega/2$. Lemma 27 then gives (62), and $\Omega_\omega \subset \mathring{\Omega}$ turns it into (16) with $c_\nu := \sqrt{n}\,\ell_\omega\|H\|_\infty$. Finally, let $t \ge T_1$, so that $S_0(x_j) \in [\omega, 1-\omega]$. By (15), $(H(y+\nu))_j = S_0(x_j) + (H\nu)_j \in [\omega/2, 1-\omega/2]$, so that $\hat{x}_j$ is well defined and $|\hat{x}_j - x_j| \le \ell_\omega\delta_H \le c_\nu\delta$. □

5 Numerical simulation

We illustrate the procedure of Theorem 16 on a two-dimensional Wilson–Cowan type network with piecewise linear activation [46, 20],

$$\begin{cases} \dot{x} = -\lambda x + CS(x) + \sigma_w \dot{w} + u, \\ y = WS(x) + \nu, \end{cases} \tag{74}$$

with $n = q = m = 2$, $\lambda = 1.2$, and C, W unknown matrices whose entries are drawn independently and uniformly in $[-1, 1]$. The saturation is componentwise, $S(x) = (S_0(x_1), S_0(x_2))^\top$, with

$$S_0(\xi) = \operatorname{clamp}\Big(\frac{\xi+2}{4},\, 0,\, 1\Big) = \begin{cases} 0 & \xi \le -2, \\ \dfrac{\xi+2}{4} & -2 < \xi < 2, \\ 1 & \xi \ge 2. \end{cases} \tag{75}$$

Here w is a standard Brownian motion and ν a zero-mean Gaussian white noise, with $\sigma_w = 0.5$ and $\sigma_\nu = 0.3$ unless stated otherwise. The system is integrated from $x(0) = (2, 4)^\top$ by an Euler–Maruyama scheme with step $T_s = 10^{-3}$, over 100 runs in which C, W and the noises are redrawn. This is (14) with $A = -\lambda \operatorname{Id}$ known and $\phi = CS(x) + \sigma_w \dot{w}$. Since the noise $\sigma_w \dot{w}$ is not bounded, the simulation probes the method slightly outside the class covered by the theorems.

Identification of W The open-loop controls are taken along $b = (1, -1)^\top$, whose span meets the regions $K_1 = [2, +\infty) \times (-\infty, -2]$ and $K_2 = (-\infty, -2] \times [2, +\infty)$ of (3), where S takes the values $v_1 = e_1$ and $v_2 = e_2$. With $\tau = 10$, $t_1 = \tau$ and $t_2 = 3\tau$, the control $u = (\mu_1 \mathbf{1}_{[0, t_1+\tau]} - \mu_2 \mathbf{1}_{[t_1+\tau, t_2+\tau]})b$, with $\mu_1 = 6$ and $\mu_2 = 7$, is the control of Lemma 26 for the design $\mathcal{E} = \{(v_1, [t_1, t_1 + \tau]), (v_2, [t_2, t_2 + \tau])\}$. Its equilibria $\mu_1 b/\lambda = (5, -5)$ and $-\mu_2 b/\lambda \simeq (-5.8, 5.8)$ lie inside K_1 and K_2 by more than the shift $1/\lambda \simeq 0.83$ due to the coupling term. Case *ii.* of Theorem 12 would allow an arbitrarily short horizon, but long windows display the averaging of the output noise by (6). Indeed, $\Gamma(\mathcal{E}) = \tau \operatorname{Id}$, which implies that the root mean square error is $2\sigma_\nu (T_s/\tau)^{1/2}$ in Frobenius norm. The bound of Remark 7, which holds for any bounded noise, does not see this averaging. The median error in Figure 3 (right) coincides perfectly with this prediction.

Tracking At $T = t_2 + \tau = 40$ the control switches to the output feedback (19), built with $\hat{W} = E(y; \mathcal{E})$ and fed with the noisy output, with $k = 10$ and $\omega = 0.1$. On $[T, T + 20]$ the reference is the circle $\gamma(t) = \big(\cos \frac{t-T}{2}, \sin \frac{t-T}{2}\big)^\top \in \Omega_{2\omega} = [-1.2, 1.2]^2$. On $[T + 5, T + 20]$ the state stayed in $\mathring{\Omega}$ in 99 of the 100 runs, and the tracking error $\|x(t) - \gamma(t)\|$, averaged over this interval, had a median of 0.33. It grows with $\|H\|_\infty = \|\hat{W}^{-1}\|_\infty$, from 0.21 for $\|H\|_\infty < 2$ to 0.60 for $\|H\|_\infty \ge 8$. This holds although ν is unbounded and k is below the gain $k^\star$ of that theorem.

Identification of C The reference then moves along a C^1 path, in 2 time units, to $p^1 = (0.8, -0.8)^\top$, dwells there for 100 time units, and moves likewise to $p^2 = -p^1$, where it dwells again. Since λ is known and $P = [S(p^1), S(p^2)]$ is invertible, this is item *iii.* of Theorem 16 with $\sigma = S$. The estimate $\hat{C} := [d_1, d_2]Z^{-1}$ of (18) is evaluated with $\hat{x} = S_0^{-1}(Hy)$ in place

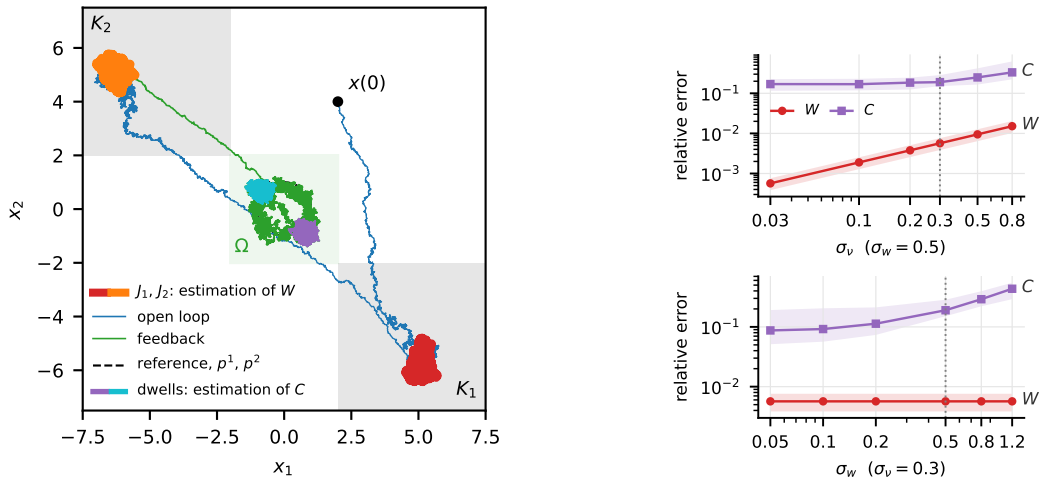


Figure 3: Left: trajectory of x in the phase plane for one run. Right: relative errors $\|\hat{W} - W\|_F/\|W\|_F$ and $\|\hat{C} - C\|_F/\|C\|_F$, median and interquartile range over the runs, against σ_ν (top) and σ_w (bottom). Dotted: the noise levels of the left panel.

of x and Hy in place of $S(x)$, where $H = \hat{W}^{-1}$ and S_0^{-1} is extended[5] affinely outside $[0, 1]$, on windows I_ℓ starting 3 time units into each dwell, with two changes against the noise. The indicator of I_ℓ is replaced by a trapezoidal weight with ramps of 5 time units, which averages the endpoint values of $\hat{x}$, and $\hat{x}$ in d_ℓ and Hy in ζ_ℓ are sampled at alternate steps, so that their errors are independent. Without noise, $\hat{C}$ coincides with C up to 10^{-5}. Figure 3 (right) shows that the error grows with both noises and is dominated by the state noise, which lies outside the class of Theorem 16. This noise also drives the state out of $\mathring{\Omega}$ at some time after $T + 5$ in 12 runs.

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